

PAPER-2

Time : 3 Hours

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **36** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble **COMPLETELY.**
12. The correct way of darkening a bubble is as :
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

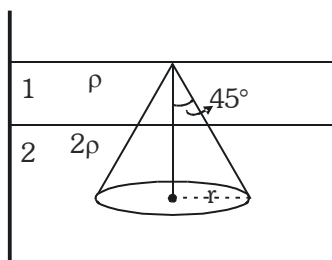
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

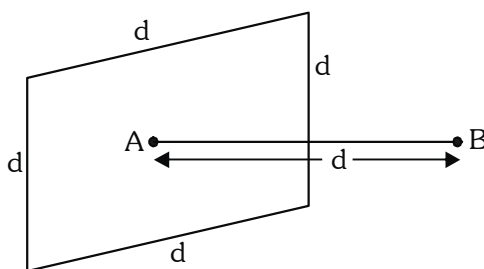
1. A solid cone of base radius 'r' and half angle 45° made of material have density $\frac{3\rho}{2}$. Cone is placed inside a container which is filled with two immiscible liquid of density ρ and 2ρ respectively. The cone is at rest with respect to container as shown in diagram. Which of the following is/are **CORRECT** statements ? (Neglect effect of atmospheric pressure)



- (A) The ratio of volume of cone in liquid 1 and liquid 2 is 1.
- (B) The force exerted by the liquid 1 on the cone in vertical direction is $\frac{\pi r^2}{3} \rho g$
- (C) Option (A) and (B) both are correct
- (D) Option (A) and (B) both are wrong.

Space for Rough Work

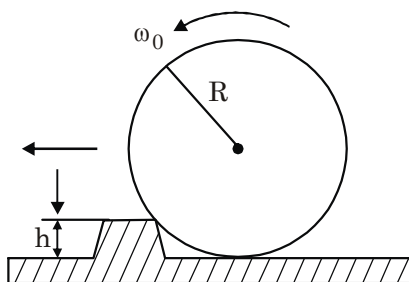
2. Consider a square frame made of uniformly charged non conducting wires of length d and linear charge density $\lambda \frac{C}{m}$. Choose the **CORRECT** statement(s).



- (A) The electric field at A is zero.
- (B) The potential at centre A is higher than potential at B.
- (C) The net electric field at B is $\left[\frac{32 K \lambda}{\sqrt{150d}} \right] \frac{N}{C}$.
- (D) The potential at B is $V = 4K\lambda \log_e \left[\frac{\sec \theta + \tan \theta}{\sec \theta - \tan \theta} \right] \frac{J}{C}$: where θ is $\sin^{-1} \left[\frac{1}{\sqrt{6}} \right]$
3. In rural areas where electric power lines carry current to big cities, it is possible to generate small amount of electricity from the lines by means of induction in a single coil. The overhead power lines carry relatively large currents that periodically reverse in direction 50 time per second. How would you orient the plane of the coil to produce maximum induced current if the power lines ran in a north-to-south direction ?
- (A) Parallel to the ground.
- (B) Perpendicular to the ground in the North-South directions.
- (C) Perpendicular to the ground in the East-West direction.
- (D) The orientation of the coil would not affect the size of the induced current.

Space for Rough Work

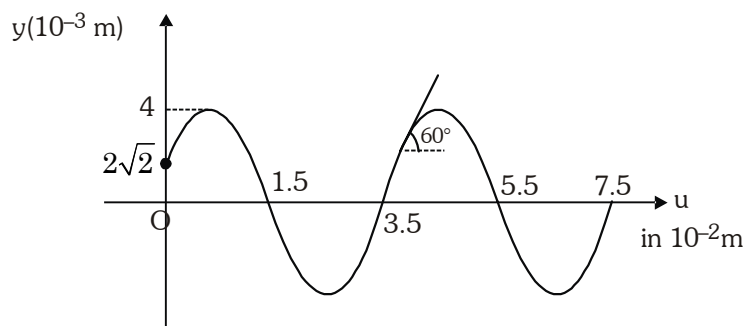
4. The engines of a spacecraft in circular orbit about the earth are fired so as to give the craft a forward thrust. How will the space craft's energy be changed after the firing, assuming another circular orbit is achieved ?
- (A) K.E. increased and P.E. decreased. (B) K.E. decreased and P.E. increased
(C) Both K.E. and P.E. increased (D) Total energy increases.
5. A solid ball of mass M and radius ' R ' is rolling without slipping on a flat horizontal surface at an initial angular velocity ω_0 . It hits a small bump of height h . Assume that the ball pivots about the tip of the bump during and after impact. Which of the following statements are **TRUE** ?



- (A) The angular velocity of the ball immediately after impact is $\omega = \omega_0 \left(1 - \frac{5h}{7R}\right)$
- (B) The minimum initial angular velocity so that the ball just makes it over the bump is $\omega_0 = \frac{7}{(7R - 5h)} \sqrt{\frac{10gh}{7}}$
- (C) The angular velocity of the ball immediately after impact is $\omega = \omega_0 \left(1 - \frac{3h}{7R}\right)$
- (D) During the impact gravitation force is considered to be impulsive.

Space for Rough Work

6. The figure shows a snapshot of a vibrating string at $t = 0$. The particle P is observed moving up with a velocity $20\sqrt{3} \text{ cm/s}$. The tangent at P makes an angle 60° with x-axis. Then (μ mass per unit length of string = 50 gm/m)



- (A) the wave is moving along negative x-axis
 (B) the wave is moving along positive x-axis
 (C) the equation of wave is given by $y = 0.4 \sin 2\pi \left(5t - \frac{x}{4} + \frac{1}{12} \right)$
 (D) the total energy carried by the wave per cycle is $3.6 \times 10^{-6} \text{ J}$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

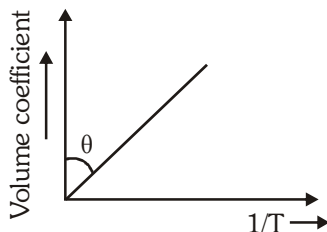
- This section contains **FOUR** questions.
- **Each question has matching lists.** The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

7. Match the following.

List-I	List-II
(P) Linear combination of two SHMs	(1) $T = 2\pi\sqrt{\frac{R}{g}}$ and not $T = \infty$
(Q) $y = A \sin \omega_1 t + A \sin (\omega_2 t + \phi)$	(2) SHM for equal frequencies and amplitude
(R) Time period of a pendulum of infinite length	(3) Superposition may not be a SHM always
(S) Maximum value of time period of an Oscillating pendulum	(4) Amplitude will be $\sqrt{2}A$ for $\omega_1 = \omega_2$ and phase difference of $\frac{\pi}{2}$
(A) $P \rightarrow 2,3$; $Q \rightarrow 2,3,4$; $R \rightarrow 1$; $S \rightarrow 1$	
(B) $P \rightarrow 1,3$; $Q \rightarrow 2,3$; $R \rightarrow 1,4$; $S \rightarrow 1$	
(C) $P \rightarrow 1,3$; $Q \rightarrow 2,3,4$; $R \rightarrow 1$; $S \rightarrow 2,3$	
(D) $P \rightarrow 2,4$; $Q \rightarrow 1,3$; $R \rightarrow 2,3$; $S \rightarrow 3,4$	

Space for Rough Work

8. A monoatomic ideal gas of 2 moles was found to be undergoing a thermodynamic polytropic process whose volume coefficient is varying with temperature as shown. ($R = \frac{25}{3} \text{ J / mol / k.}$)



List-I

- (P) Work done by gas in joules for every 1K rise in temperature, if $\tan \theta = 3$
- (Q) Internal energy rise of the gas in joules for every 1K rise in temperature, if $\tan \theta = 2$
- (R) Heat absorbed by the gas in joules for every 1K rise in temperature, if $\tan \theta = 1$
- (S) Value of θ in degrees for which the internal energy rise of the gas for every 1 K rise in temperature is 25 J

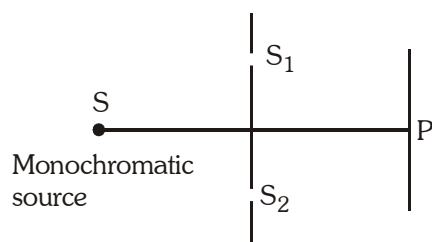
List-II

- (1) 0
- (2) 25
- (3) $\frac{125}{3}$
- (4) $\frac{50}{9}$

- (A) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 1,2,3,4$
- (B) $P \rightarrow 2,3; Q \rightarrow 1,3,4; R \rightarrow 2; S \rightarrow 3,4$
- (C) $P \rightarrow 3; Q \rightarrow 3,4; R \rightarrow 2,3,4; S \rightarrow 1,2$
- (D) $P \rightarrow 2,4; Q \rightarrow 1,2; R \rightarrow 4; S \rightarrow 1,3,4$

Space for Rough Work

9. A Young's double slit apparatus is shown in figure



List-I

- (P) Source S move upwards
 (Q) A thin glass slab is placed in front of slit S_2
 (R) Whole apparatus is immersed in water
 (S) Diameter of slit S_2 is doubled

List-II

- (1) Central fringe moves upwards
 (2) Central fringe moves downward
 (3) Fringe width remains constant
 (4) Fringe width decreases
 (5) Intensity of central maxima increases

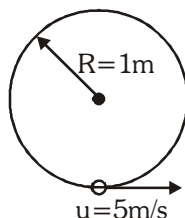
- (A) $P \rightarrow 1,2$; $Q \rightarrow 3,4$; $R \rightarrow 2,5$; $S \rightarrow 4$
 (B) $P \rightarrow 2,3$; $Q \rightarrow 2,3$; $R \rightarrow 4$; $S \rightarrow 3,5$
 (C) $P \rightarrow 2,3$; $Q \rightarrow 3,4$; $R \rightarrow 1,4$; $S \rightarrow 1,2$
 (D) $P \rightarrow 1,2$; $Q \rightarrow 2,3$; $R \rightarrow 2,4$; $S \rightarrow 1,3$

Space for Rough Work

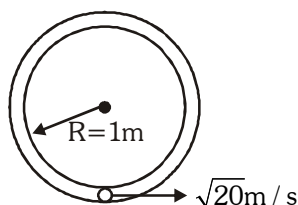
10. In list-I a situation is depicted each of which is in vertical plane. The surfaces are frictionless. Match these with appropriate entries in list-II.

List-I

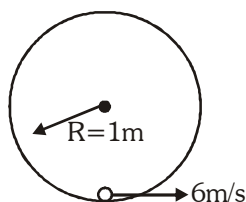
- (P) Bead is threaded on a circular fixed wire and is projected from the lowest point



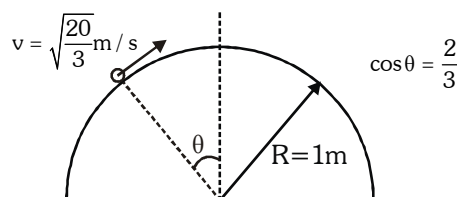
- (Q) Block loosely fits inside the fixed small tube and is projected from the lowest point



- (R) Block is projected horizontally from the lowest point of a smooth fixed cylinder



- (S) Block is projected on a fixed hemisphere from angular position θ



List-II

- (1) Normal force is zero. At the top most point of its trajectory

- (2) Velocity of the body is zero. At top most point of it's trajectory

- (3) Acceleration of the body is zero at the top most point of it's trajectory

- (4) Normal force is radially outwards at the top most point of trajectory

- (A) P $\rightarrow$ 1,2; Q $\rightarrow$ 3,4; R $\rightarrow$ 2,5; S $\rightarrow$ 1,3

- (B) P $\rightarrow$ 2,3; Q $\rightarrow$ 2,3,4; R $\rightarrow$ 3,4; S $\rightarrow$ 1,2

- (C) P $\rightarrow$ 2,3; Q $\rightarrow$ 3,4; R $\rightarrow$ 2,3,4; S $\rightarrow$ 1,2

- (D) P $\rightarrow$ 2,4; Q $\rightarrow$ 1,2; R $\rightarrow$ 1; S $\rightarrow$ 2,3,4

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

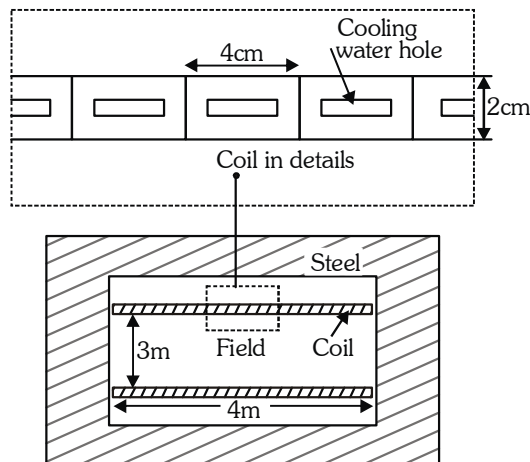
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

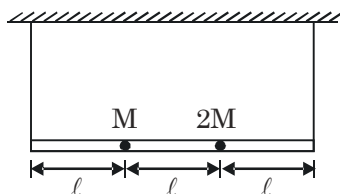
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. A large solenoid coil is made of single layer of conductor of cross section $4\text{cm} \times 2\text{cm}$ with a cooling water hole $2\text{cm} \times 1\text{cm}$ in the conductor. The coil, which consists of 100 turns, has a diameter of 3 meters, and a length of 4 meters. At the two ends of coil are circular steel plates to make the field uniform and to return the magnetic flux through a steel cylindrical structure to the coil, as shown in the diagram. A magnetic field of 0.25 tesla is desired. Resistivity of conductor is 3×10^{-8} ohm-meter. What power (in megawatt) must be supplied to provide the desired field ?



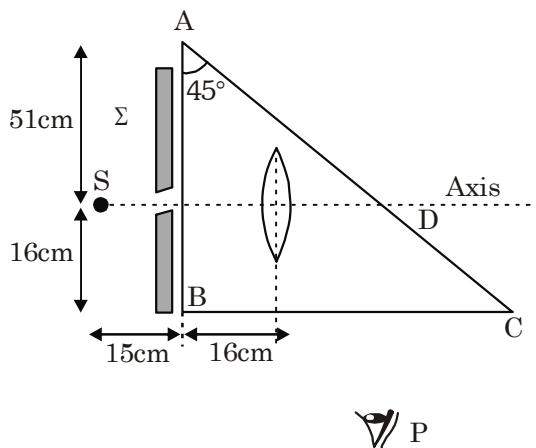
Space for Rough Work

2. To determine the density of blood, drops of blood were added to a mixture of xylene with a density of $0.867 \frac{\text{gm}}{\text{cm}^3}$, and bromobenzene with a density of $1.497 \frac{\text{g}}{\text{cm}^3}$. The ratio of xylene and bromobenzene was being changed until the drops were floating. i.e. they did not rise nor sink. This happened when the mixture contained 72% of xylene and 28% of bromobenzene by volume. What was the density (in gm/cm^3) of blood?
3. A 68g bungee cord has an equilibrium length of 1.30m. The cord is stretched to a length of 1.80m, then vibrated at 20 Hz. This produces a standing wave with two antinodes. What is the spring constant (in N/m) of bungee cord ? Assume both ends fixed?
4. A rod with length 3ℓ and insignificant mass is suspended in horizontal position using two fibers attached to ends of the rod. There are two masses of M and $2M$ fixed to the rod, so their centres are placed in one third or two thirds of the rod's length. Find the ratio of tension force in uncut string after and before cutting of string close to $2M$ will be :



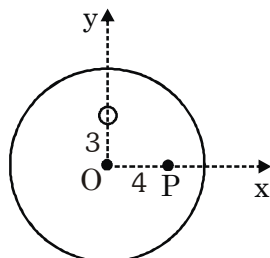
Space for Rough Work

5. Wire of an original length of 10 m is fixed at one end and is being stretched with force of 200 N which elongates it by 4mm. Find the difference in diameter as multiple of 10^{-7} during the stretching of the wire. Where, Young's modulus $Y = 2 \times 10^{11} \text{ N-m}^2$ and poisson's ratio = 0.32
6. A conductor of length $\ell = 10 \text{ cm}$ is placed in E-W direction in a plane. Earth's horizontal magnetic field is $B = 3 \text{ Tesla}$. Find the amount of charge (in mC) which passes through it when it is found to jump to height 5 cm. Mass of rod is 2 gm and $g = 10 \text{ m/s}^2$.
7. An isosceles right angled triangular glass ($\mu = 1.6$) prism has a cavity inside it in the shape of a thin convex lens whose both surface have radius of curvature equal to 20 cm. The cavity has been filled with a transparent liquid of refractive index 2.4. S is a point object and there is an opaque sheet having a small hole such that the source and the hole both lie on the principal axis of the lens. The small hole in the opaque sheet is just to ensure that only paraxial rays are incident on the original system. However, the size of hole is large enough to neglect diffraction effects. At what perpendicular distance (in cm) from line BC, the observer P will see the image?



Space for Rough Work

8. A non-conducting sphere of radius $R = 5\text{cm}$ has its center at origin O of co-ordinate system shown in figure. It has a spherical cavity of radius $r = 1\text{ cm}$, whose centre is at $(0, 3\text{cm})$. Solid material of sphere has uniform positive charge density $\rho = \frac{10^{-6}}{\pi} \text{ coul. m}^{-3}$. Assume dielectric constant of material to be 1. The potential (in volt) at point $P (4\text{cm}, 0)$ is



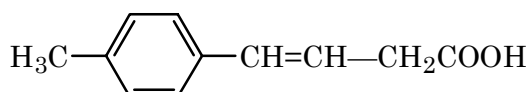
Space for Rough Work

PART-2 : CHEMISTRY

SECTION-I(i) : (Maximum Marks: 24)

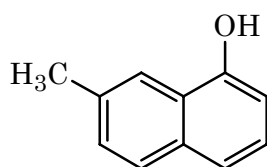
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1. In a chemical reaction, the compound (X)



4(p-tolyl)-3 butenoic acid

Can rather conveniently be converted to another compound (Y) which is

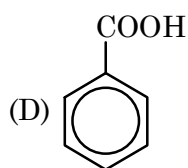
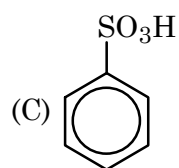
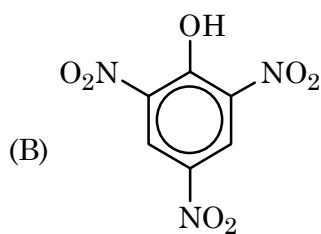
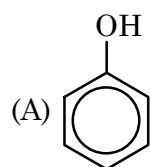
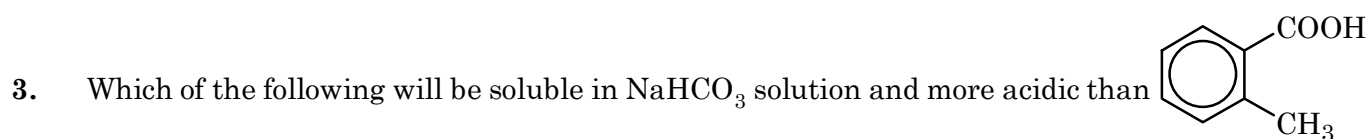
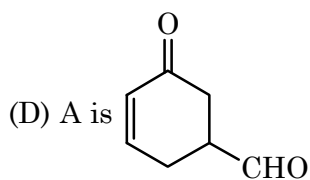
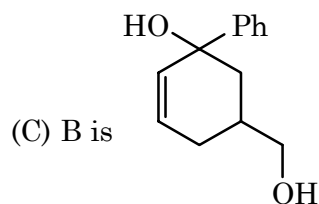
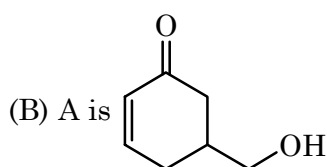
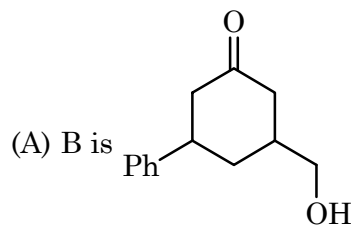
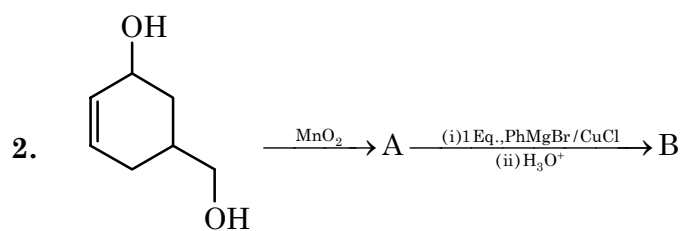


7-methyl 1-naphthol

Which of the following is/are suitable for this purpose.

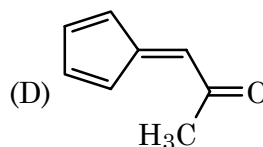
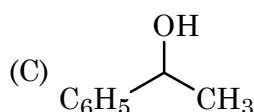
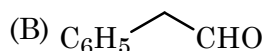
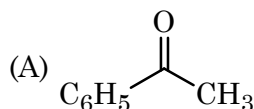
- (A) Trans isomer of (X) can be treated with anhydrous AlCl_3 .
- (B) Cis isomer of (X) can be treated with HF.
- (C) Trans isomer of (X) can be treated with conc. H_3PO_4 .
- (D) Cis isomer of (X) can be treated with NaOH.

Space for Rough Work



Space for Rough Work

4. Which statements is/are correct among the following ?
 (A) BF_3 is strongest Lewis acid among BF_3 , BCl_3 , BBr_3
 (B) Between NH_3 and PH_3 , NH_3 is better electron donor because NH_3 has more directional lone pair.
 (C) In P_4O_{10} there are 5, P–O–P bonds.
 (D) ClOF_3 has same shape as XeO_2F_2 .
5. Compound (X), molecular formula $\text{C}_8\text{H}_8\text{O}$, is water insoluble and gives yellow ppt. with I_2/NaOH . It reacts with $\text{H}_2\text{N}-\text{OH}$ in the presence of $\text{NH}_4\text{Cl}/\text{NH}_4\text{OH}$ to give two geometrical isomers, which can go for rearrangement with acids. Compound (X) can be,



6. In order to study effect of temperature on rate of reaction, another term “ σ ” is defined as

$\sigma = \frac{1}{k} \frac{dk}{dT}$ [where k is rate constant and T represents temperature]. If for a reaction, the value

of $\sigma = \frac{1.25 \times 10^6}{T^3} \text{ K}^{-1}$ [T should be in Kelvin], then identify the correct statement(s).

- (A) Activation energy for the given reaction at 250 Kelvin is 10 Kcal/mol.
 (B) Activation energy for the given reaction at 2000 Kelvin is 1.25 Kcal/mol.
 (C) Activation energy of this reaction increases on increasing temperature.
 (D) The value of $\frac{dk}{dT}$ at 2000 Kelvin is $0.625 \times k$.

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- **Each question has matching lists.** The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

7. Match the following.

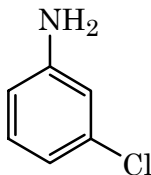
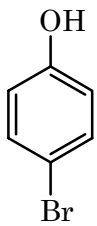
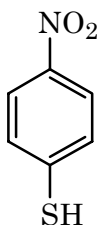
List - I (Conversion)	List - II (Reagent)
(P) $\text{R}-\overset{\text{O}}{\underset{\parallel}{\text{C}}}-\text{NH}_2 \longrightarrow \text{RNH}_2$	(1) KBr
(Q) $\text{R}-\overset{\text{O}}{\underset{\parallel}{\text{C}}}-\text{NH}-\text{O}-\overset{\text{O}}{\underset{\parallel}{\text{C}}}-\text{R} \longrightarrow \text{RNH}_2$	(2) $\text{OH}^-/\text{H}_2\text{O}$
(R) $\text{R}-\overset{\text{O}}{\underset{\parallel}{\text{C}}}-\text{OH} \longrightarrow \text{RNH}_2$	(3) $\text{N}_3\text{H}/\text{H}^+$
(S) $\text{R}-\overset{\text{O}}{\underset{\parallel}{\text{C}}}-\text{NH}-\text{NH}_2 \longrightarrow \text{RNH}_2$	(4) $\text{HNO}_2/\Delta/\text{H}_3\text{O}^+$

Codes :

- | | | | |
|-----------|-------|-------|-------|
| (A) P→(1) | Q→(3) | R→(2) | S→(4) |
| (B) P→(1) | Q→(2) | R→(3) | S→(1) |
| (C) P→(1) | Q→(2) | R→(3) | S→(4) |
| (D) P→(2) | Q→(3) | R→(4) | S→(1) |

Space for Rough Work

8. Match the following.

List - I (Organic compound)	List - II (Estimation/Detection Possible using)
(P) 	(1) Dumas method
(Q) 	(2) Kjeldahl's method
(R) 	(3) Carius method
(S) $C_2H_5NH_2$	(4) $p-CH_3C_6H_4SO_2Cl$ (Hinsberg method)

Codes :

- | | | | |
|-----------------|---------|-----------|-----------|
| (A) P→(1,2,3) | Q→(3,4) | R→(1,2) | S→(1,2,4) |
| (B) P→(1,2,3,4) | Q→(3) | R→(1,2,3) | S→(1,2,4) |
| (C) P→(1,2,3) | Q→(1,4) | R→(2,3) | S→(4) |
| (D) P→(1,4) | Q→(3,4) | R→(2,3) | S→(1,3,4) |

Space for Rough Work

9. Match the following.

A = Central atom

X = Neighbouring atom(s)

List - I	List - II
(P) PCl_3F_2 & SOF_4	(1) $\angle\text{X-A-X} = 120^\circ$ observed in both species
(Q) BF_3 & BH_3	(2) $\angle\text{X-A-X}$ less than 120° observed in both species
(R) NO_3^- & NF_3	(3) Lone pair(s) at 'A' in both species
(S) XeF_5^+ & ICl_2^+	(4) Both species non-polar
	(5) Same 'A' but different hybridization in both species

Codes :

- | | | | |
|-------------|-----------|-------|---------|
| (A) P→(1) | Q→(4,5) | R→(5) | S→(3) |
| (B) P→(1,2) | Q→(1,4) | R→(5) | S→(3) |
| (C) P→(2) | Q→(1,4,5) | R→(5) | S→(2,3) |
| (D) P→(2) | Q→(1,4) | R→(5) | S→(2,3) |

Space for Rough Work

10. Match the following.

List - I Electronic transition of hydrogenic "sample"	List - II Observation for transition(s)
(P) $n = 4$ to $n = 2$ for He^+	(1) Total 10 emission lines
(Q) $n = 5$ to $n = 2$ for H	(2) "Balmer" line(s) in visible spectrum
(R) $n = 6$ to $n = 3$ for Li^{2+}	(3) "Balmer" line(s) in U-V spectrum
(S) $n = 8$ to $n = 4$ for H	(4) Total 6 emission lines
	(5) No "Paschen" line(s) observed

Codes :

- (A) $P \rightarrow (2)$ $Q \rightarrow (2,4)$ $R \rightarrow (4)$ $S \rightarrow (5)$
 (B) $P \rightarrow (3)$ $Q \rightarrow (2)$ $R \rightarrow (4)$ $S \rightarrow (4,5)$
 (C) $P \rightarrow (3)$ $Q \rightarrow (2,4)$ $R \rightarrow (4)$ $S \rightarrow (1,5)$
 (D) $P \rightarrow (2)$ $Q \rightarrow (2,4)$ $R \rightarrow (4,5)$ $S \rightarrow (1)$

 Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	●	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	●	●	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	0	0	●	●	●	●	●
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	●	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If ONLY the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. A 10.0 gram sample of a mixture of calcium chloride and sodium chloride is treated with Na_2CO_3 . The precipitate formed is decomposed on heating. The weight of the residue is 1.68 gram. The percentage by mass of CaCl_2 in the original mixture is :

2. The emf of the following cell is given as $E_{\text{Cell}} = -a \times 10^{-1} \text{ V}$
 $\text{Pt}, \text{H}_2(1 \text{ atm}) | \text{CH}_3\text{COOH}(0.1 \text{ M}) || \text{NH}_4\text{OH}(0.1 \text{ M}) | \text{H}_2(1 \text{ atm}), \text{Pt}$

$$K_a \text{ for } \text{CH}_3\text{COOH} = 10^{-5}; K_b \text{ for } \text{NH}_4\text{OH} = 10^{-5}, \text{ Given } \frac{2.303RT}{F} = 0.06 \text{ V}$$

What is the value of a ?

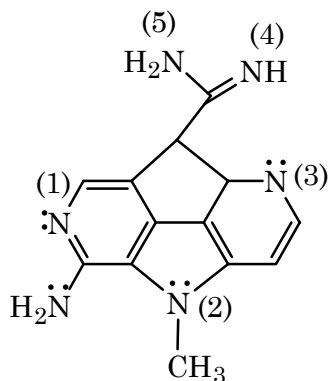
3. Consider the following two first order reaction :



Reaction (i) is 75% complete in 4 hrs. while reaction (ii) takes 16 hrs. for same 75% completion of reaction under identical conditions. By how many times, half life of (ii) is greater than the half life of (i) ?

Space for Rough Work

4. A mixture contains AgCl , $\text{Al}(\text{OH})_3$, $\text{Zn}(\text{OH})_2$, $\text{Cu}(\text{OH})_2$, $\text{Fe}(\text{OH})_3$ & $\text{Cd}(\text{OH})_2$. On adding excess of NH_4OH how many of the metal complex cations will transfer into filtrate ?
(Assume original mixture do not react among themselves)
5. Sum of tags of Nitrogen lone pairs which are delocalized.



6. Identify the species in which the "actual" as well as the "average" Oxidation number of the marked element is/are an EVEN integer

- | | | | |
|--|--|---|---|
| (1) $\text{H}_4\underline{\text{P}}_2\text{O}_6$ | (2) $\text{Na}_2 \underline{\text{S}}_4\text{O}_6$ | (3) $\text{H}_2\underline{\text{S}} \text{O}_5$ | (4) $\text{H}_2 \underline{\text{S}}_2\text{O}_8$ |
| (5) $\text{H}_2\underline{\text{S}}_2\text{O}_7$ | (6) $\underline{\text{S}}_8$ | (7) $\underline{\text{Cr}}\text{O}_8^{3-}$ | (8) $\text{H}\underline{\text{N}}\text{O}_4$ |
| (9) $\underline{\text{N}}_2\text{O}_2^{2-}$ | (10) $\underline{\text{Os}}\text{O}_4$ | (11) $\underline{\text{P}}\text{H}_3$ | (12) $\underline{\text{Cr}}\text{O}_4^{2-}$ |

Space for Rough Work

7. A solution is created by mixing equal volumes of 0.1 M Na_2SO_4 and 0.1 M BaCl_2 . The resultant solution volume is 1 litre, its temperature rises to $T = 50^\circ\text{C}$ and it also contains a precipitate mass. Through highly accurate measurements, total ionic concentration is found to be 0.20005 M (excluding H^+ and OH^- ions). Then report the number of correct statement(s) among the following
- (A) Final solution contains equal moles of all ions (excluding H^+ and OH^- ions)
 - (B) Final solution contains equal moles of Na^+ and Cl^- ions.
 - (C) Final solution contains equal moles of Ba^{+2} and SO_4^{2-} ions.
 - (D) K_{sp} value of BaSO_4 is 2.5×10^{-9} at temperature T
 - (E) pH of resultant solution is 7
 - (F) Resultant solution after filtration can act as buffer solution
 - (G) Final solution is neutral because has $[\text{H}^+]_{\text{eqm}} = [\text{OH}^-]_{\text{eqm}}$
8. 40 ml of 0.05 M solution of sodium sesquicarbonate ($\text{Na}_2\text{CO}_3 \cdot \text{NaHCO}_3 \cdot 2\text{H}_2\text{O}$) is titrated against 0.05 M HCl. "X" ml of HCl is used when phenolphthalein is the indicator and "Y" ml of HCl is used when methyl orange is the indicator in two separate titrations. Hence, value of $\frac{(Y - X)}{10}$ is

Space for Rough Work

PART-3 : MATHEMATICS

SECTION-I(i) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. If $a\ell^2 + bm^2 + 2d\ell + 1 = 0$, where a, b, d are fixed real numbers such that $a - b = d^2$, then the line $\ell x + my + 1 = 0$ touches the fixed circle (where $b < 0$ and $\ell, m \in \mathbb{R}$)
 - (A) which cut the x-axis orthogonally
 - (B) with radius equal to $\sqrt{-b}$
 - (C) on which length of the tangent from origin is $\sqrt{d^2 + b}$
 - (D) with centre $\equiv (d, -b)$
2. Let p and q are complex coefficient of quadratic trinomial $p(x) = x^2 + px + q$. It is given that $p(p(1)) = p(p(2)) = 0$ and $p(1) \neq p(2)$, then which of the following is/are true.

(A) $ p(0) = \frac{1}{2}$	(B) $p(0) = \frac{-3}{2}$
(C) $ p(1) = 1$	(D) $p(x) = 0$ has real roots

Space for Rough Work

3. A differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies following relations

$$(i) f(x) - f(y) = (x - y)f'\left(\frac{x+y}{2}\right), \forall x, y \in \mathbb{R} \quad (ii) f(0) = f(1) = 0, f(2) > 0$$

which of the following is/are true

(A) $f(x)$ is an onto function

(B) $f(x)$ has an extrema at $x = \frac{1}{2}$

(C) $f(x)$ is a many one function

(D) $\lim_{x \rightarrow -\infty} f(x) = -\infty$

4. If A, B are square matrix of order n, then which of the following is/are always true

(A) If A, B and $A + B$ are idempotent matrices then $AB = BA = O$

(B) If A and B are orthogonal matrices then AB and BA are also orthogonal matrices

(C) If $A + B = I$ and A is idempotent matrix then $AB = BA = O$

(D) If $A^2 + B^2 = AB$ then $\text{tr}(ABAB) = \text{tr}(A^2B^2)$

5. Let $P = |z^5 + \bar{z}^3 + 6z| - 2|z^4 + 1|$ and $|z| = 1$ and Q is number of real roots of $5 - x = \sqrt{x + \sqrt{x + \sqrt{x}}}$, where $z \in \mathbb{C}$, $x \in \mathbb{R}$ then which of the following is/are FALSE ?

(A) $P_{\min} + Q = 5$

(B) $P_{\max} + Q = 4$

(C) $P + Q \in [4, 5]$

(D) $P - Q \in [3, 4]$

(where $P_{\min}$ and $P_{\max}$ are minimum and maximum value of P respectively)

Space for Rough Work

6. Identify the correct statements

(A) If $g(x)$ is a function satisfying $g(x) + g(x + 2) = 10, \forall x \in \mathbb{R}$ then $\int_{-1}^7 g(x) dx = 40$

(B) $\lim_{\theta \rightarrow 0^+} \int_0^{\infty} \frac{\tan^{-1} x}{x^2 + 2x \cos \theta + 1} dx = \frac{\pi}{4}$

(C) If $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_{n-1}$ are the imaginary n th roots of unity then $\left| \prod_{r=1}^{n-1} (i - \alpha_r) \right|$ (where $i = \sqrt{-1}$) can take the value equal to $\sqrt{2}$.

(D) If a, b, c are the sides opposite to the angles A, B, C respectively in a triangle ABC ,

then $\begin{vmatrix} a^2 & b \sin A & c \sin A \\ b \sin A & 1 & \cos A \\ c \sin A & \cos A & 1 \end{vmatrix}$ is equal to 1.

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- **Each question has matching lists.** The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

7. Match List-I with List-II and select the correct answer using the code given below the list.

List-I

List-II

(P) If $\lim_{n \rightarrow \infty} \left(n^{-\frac{1}{2}(1+\frac{1}{n})} (1^1 2^2 3^3 4^4 \dots n^n)^{\frac{1}{n^2}} \right) = e^{-\left(\frac{p}{q}\right)}$ then the value of p/q is greater than (1) 5

(Q) If $(7!)!$ is divisible by $(7!)^{k!} \cdot (6!)!$ then the maximum value of k is (2) 6

(R) The value of $\lim_{x \rightarrow 0^+} x^{(x^x-1)}$ is less than or equal to (3) $\frac{1}{5}$

(S) If α, β, γ are the roots of the equation $x^3 - 2x^2 - 1 = 0$ and

$T_n = \alpha^n + \beta^n + \gamma^n$ then the value of $\frac{T_{11} - T_8}{T_{10}}$ is greater than (4) $\frac{1}{6}$

(5) $\frac{1}{4}$

(A) P $\rightarrow$ 3,4; Q $\rightarrow$ 2,4; R $\rightarrow$ 1; S $\rightarrow$ 4,5

(B) P $\rightarrow$ 1,4; Q $\rightarrow$ 2; R $\rightarrow$ 1; S $\rightarrow$ 3,4

(C) P $\rightarrow$ 3; Q $\rightarrow$ 2,4; R $\rightarrow$ 4; S $\rightarrow$ 1,5

(D) P $\rightarrow$ 3,4; Q $\rightarrow$ 2; R $\rightarrow$ 1,2; S $\rightarrow$ 3,4,5

Space for Rough Work

8. Match List-I with List-II and select the correct answer using the code given below the list.

List-I

List-II

- (P) If the exhaustive set of x satisfying $[5x] + [4x] + [3x] = 2$ is $[a, b)$ then an integer belongs to $[2^3a, 3^3b)$ is $([.] \text{ is GIF})$ (1) 3
- (Q) Number of critical points of $f(x) = \max\{x^3 - 1, x^2 - 1\}$ is less than or equal to (2) 5
- (R) For the function $f(x) = \begin{cases} \frac{\{x\} - \{-x\}}{\{x\} + \{-x\}}, & x \notin \mathbb{Z} \\ \cos \pi x, & x \in \mathbb{Z} \end{cases}$ then number points of discontinuity of $f(x)$ in $x \in [-2, 2]$ (3) 8
- (S) The value of x for which $(\sin^{-1} x)^2 - (\cos^{-1} x)^2 \geq \frac{\pi^2}{4}$ is always less than or equal to (4) 1
- (5) 4

- (A) $P \rightarrow 1, 2, 3; Q \rightarrow 2, 4; R \rightarrow 1; S \rightarrow 4, 5$
- (B) $P \rightarrow 1, 2, 3, 5; Q \rightarrow 1, 2, 3, 5; R \rightarrow 5; S \rightarrow 1, 2, 3, 4, 5$
- (C) $P \rightarrow 1, 2, 3, 5; Q \rightarrow 1, 2, 3; R \rightarrow 5; S \rightarrow 1, 2, 3, 5$
- (D) $P \rightarrow 1, 3, 5; Q \rightarrow 1, 2, 3; R \rightarrow 1; S \rightarrow 1, 2, 3, 4$

Space for Rough Work

9. Match List-I with List-II and select the correct answer using the code given below the list.

List-I

List-II

(P) Let p, q be real numbers such that $p^2 + q > q^2 + p$.

(1) 3

Then value $p^2 + q$ is greater than

(Q) If equation $2t^3 - 9t^2 + 30 - a = 0$ has three real and distinct roots, then possible integral values of a is/are

(2) 5

(R) Let $f : [3, \infty) \rightarrow [1, \infty)$ be defined by $f(x) = \pi^{x(x-3)}$. Number of solutions of $f(x) = f^{-1}(x)$ is always greater than

(3) $-\frac{1}{2}$

(S) Sum of series $\sum_{r=1}^{\infty} \sin^{-1} \left[\frac{2r+1}{r(r+1)(\sqrt{r^2+2r} + \sqrt{r^2-1})} \right]$ is always

(4) -1

less than

(5) 8

(A) $P \rightarrow 1, 2, 3$; $Q \rightarrow 2, 4$; $R \rightarrow 1$; $S \rightarrow 4, 5$

(B) $P \rightarrow 1, 2, 3$; $Q \rightarrow 1, 3$; $R \rightarrow 5$; $S \rightarrow 2, 5$

(C) $P \rightarrow 1, 2, 3, 4$; $Q \rightarrow 1, 2, 3$; $R \rightarrow 5$; $S \rightarrow 1, 2, 3, 5$

(D) $P \rightarrow 3, 4$; $Q \rightarrow 2, 5$; $R \rightarrow 3, 4$; $S \rightarrow 1, 2, 5$

Space for Rough Work

10. Match List-I with List-II and select the correct answer using the code given below the list.

List-I

List-II

- (P) In $\triangle ABC$, if I is incentre then $AI + BI + CI \geq \lambda r$ then λ may be (where AI, BI, CI, are distances of I from vertices and r is in radius) (1) 2
- (Q) Let $f(x) = 30 - 2x - x^3$ and $f_n(x) = \underbrace{f \circ f \circ \dots \circ f}_n(x)$, then positive integral values of x which satisfies $f_{2020}(x) < f_{2019}(-x)$, is (2) 3
- (R) The power of $x^2 + xy + y^2$ by which the polynomial $(x + y)^7 - x^7 - y^7$ is divisible, is/are (3) 5
- (S) Let a, b, c be complex such that $ab + bc + ca = \frac{1}{2}, (a + b + c) = 2, abc = 4$, then the value of $\left| 9 \left(\frac{1}{ab + c - 1} + \frac{1}{ac + b - 1} + \frac{1}{bc + a - 1} \right) \right|$ is always less than (4) 6
- (5) 1
- (A) $P \rightarrow 1, 2, 3; Q \rightarrow 2, 4; R \rightarrow 1; S \rightarrow 4, 5$
- (B) $P \rightarrow 1, 2, 3, 4, 5; Q \rightarrow 1, 5; R \rightarrow 1, 5; S \rightarrow 2, 3, 4$
- (C) $P \rightarrow 1, 2, 3, 4; Q \rightarrow 1, 2, 3; R \rightarrow 5; S \rightarrow 1, 2, 3, 5$
- (D) $P \rightarrow 3, 4; Q \rightarrow 2, 5; R \rightarrow 3, 4; S \rightarrow 1, 2, 5$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- Let $a_1, a_2, a_3 \dots a_n$ ($n \in \mathbb{N}$) is a sequence such that $a_1 = a_2 = 2$ and $\frac{2a_{n-1}a_n}{a_{n-1}a_{n+1} - a_n^2} = n^3 - n$,
 $\forall n = 2, 3, 4, \dots$, find the value of $\frac{[S]}{16}$, where $S = \sum_{k=2}^{2020} \frac{a_{k+1}}{a_k}$ and $[.]$ denotes G.I.F.)
- If f and g are two real valued differentiable functions such that
 $f(x) = x^2 + g'(0)x + g''(1)$ and
 $g(y) = y^2 + f'(y) + f'(y) - 2$ then
 $f(3) + \frac{1}{g(2)}$ is equal to
- Let $S_n = \frac{1}{1^n} + \frac{1}{3^n} + \frac{1}{5^n} + \dots \infty$ and $\int_0^1 \frac{x(\ln x)^2}{1-x^4} dx = mS_3$ then value m is

Space for Rough Work

4. Let $M = 3a7b$ and $N = a37b$ are two natural numbers, where $a, b \in \{0, 1, 2, 3, \dots, 9\}$ such that $\frac{M}{8}$ OR $\frac{N}{12}$ are integers. If the probability that $\frac{M}{12}$ AND $\frac{N}{8}$ are also integers, is p then $4p = ?$
5. From a variable point $P(t^2, 2t)$ ($1 \leq t \leq 3$) on the parabola $y^2 = 4x$, perpendicular PM is drawn to the tangent at the vertex of the parabola. Now from the midpoint Q of PM , perpendicular QL is drawn to the focal chord of parabola through P . Find the maximum length of QL
6. If $z_0 = \frac{1-i}{2}$ then the value of the product $\lim_{n \rightarrow \infty} \left| (1+z_0)(1+z_0^2)(1+z_0^4) \dots (1+z_0^{2^n}) \right|, n \in \mathbb{N}$ must be

Space for Rough Work

7. The gamma function $\Gamma(x)$ is a continuous smooth function in practice that connects the points (x, y) given by $y = (x - 1)!$ at the positive integer values for x . It is uniquely defined by $\Gamma(x+1) = x\Gamma(x)$ together with $\Gamma(1)=1$ and $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$. Another important function is beta function $B(x, y) = \int_0^1 t^{x-1} (1-t)^{y-1} dt = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$. If the value of $\int_0^{\pi/2} \sin^6 x \cdot \cos^8 x dx = \frac{5\pi}{2^b}$ then b is equal to
8. The plane $ax + by + cz = 1$ containing the line $\frac{x-3}{2} = \frac{y-1}{4} = \frac{z-2}{5}$ is rotated through an angle $\frac{\pi}{2}$ about this line to contain the point $(4, 3, 7)$. Value of $\left(2a + 3b + \frac{c}{8}\right)$ equals

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	One or more correct option(s)	6	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-2 In all other cases	24
I(ii)	Matching Lists Type (Single correct option)	4	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	-1 In all other cases	12
II	Numerical Value Type (Up to second decimal place)	8	+3 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	24

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator